

DESI DR2 BAO Alone: An Independent, Preregistered Reproduction of the Flat- Λ CDM and Flat- w CDM Constraints, a DR1→DR2 Consistency Check, and a Kernel-Verified Model Skeleton

AutoevolveAI / ANSE Research Pipeline
Model-generated with Claude Code (automated pipeline)

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Abstract

We re-derive the BAO-only cosmological constraints of the DESI DR2 release [1] from its public Gaussian BAO likelihood (13 points, full covariance). The targets, tolerances, estimator and every control were written in a preregistration and frozen before any fit (sha256 prefix 723cd27d3e5e198e). The primary estimator is the posterior mean under DESI’s flat priors, sampled with `emcee` 3.1.6. We obtain $\Omega_m = 0.29781 \pm 0.00856$ and $hr_d = 101.530 \pm 0.732$ Mpc in flat Λ CDM, and $\Omega_m = 0.29713 \pm 0.00896$, $w = -0.9164 \pm 0.0787$ in flat w CDM. The largest deviation from DESI’s quoted values is 0.037 published standard deviations, and every width agrees to within 0.9%.

An independent grid posterior agrees with the MCMC to 0.0094 MCMC standard deviations. The minimum is $\chi^2_{\min} = 10.271$ on 11 degrees of freedom (DESI: 10.2), and the Ω_m – hr_d correlation is $r = -0.9230$ (DESI: -0.92). All five positive controls behave as required, and all five negative controls fail as required. All 14 preregistered criteria pass.

Running DR1 through the same pipeline gives a shift $\Delta\Omega_m = +0.00283$, which is 0.231σ under an ideal-nesting error. A CAMB 2.0.4 cross-check and a radiation variant bound the error from omitting radiation and neutrino mass in our primary model: no parameter moves by more than 0.046σ .

Six Lean 4 theorems about the fitted w CDM background are kernel-checked with exactly the trusted axioms: positivity of $E(z)$ over the full w prior, exact nesting with Λ CDM at $w = -1$, and strict monotonicity of the comoving-distance integral. This is a reproduction, not a new cosmological result.

1 Introduction

A measurement of the baryon acoustic oscillation (BAO) feature at redshift z constrains two ratios: the transverse comoving distance $D_M(z)/r_d$ and the Hubble distance $D_H(z)/r_d = c/(H(z)r_d)$, where r_d is the sound horizon at the drag epoch. DESI DR2 [1] reports these ratios in seven redshift bins. From them alone, it quotes the following for flat Λ CDM (its eq. 17):

$$\Omega_m = 0.2975 \pm 0.0086, \quad hr_d = (101.54 \pm 0.73) \text{ Mpc}, \quad r = -0.92,$$

For flat w CDM (its Table V, row “DESI”) it quotes $\Omega_m = 0.2969 \pm 0.0089$ and $w = -0.916 \pm 0.078$.

An earlier exercise in this pipeline re-fit the DR1 release [2]. That exercise compared a χ^2 minimum with DESI’s posterior mean. At DR1 precision the difference is harmless, but it is the wrong estimator. Here we match the estimator DESI states it uses. DR2 Sec. V says: “For 1D marginalized posterior results we quote the mean and standard deviation when the distributions are symmetric.” We also extend the model to w CDM, and we measure the DR1→DR2 shift with the same code.

The whole exercise is preregistered, and the machine-readable record lives in the accompanying repository:

- `results/desi_dr2_bao/preregistration.json`: the targets, tolerances, estimator and every control;
- `results/desi_dr2_bao/fit.json`: every number in this paper (sha256 prefix 4380af6d0e1ed743);
- `results/desi_dr2_bao/ledger/`: the tier-capped claim ledger.

2 Data

We use `desi_gaussian_bao_ALL_GCcomb_{mean,cov}.txt` from the DESI DR2 BAO release. The local data README attributes it to [1]. Its sha256 digests are fixed in the preregistration and were checked again at fit time and when this paper’s tables were generated. Table 1 is produced directly from these files.

The file means match Table IV of [1] (v3) to its rounding, and $\sqrt{\text{diag } C}$ matches the tabulated errors to within about 1.5%. There is one caveat. The file’s D_M – D_H correlation coefficients differ from Table IV for some bins. For LRG3+ELG1 the file gives -0.347 , while Table IV gives -0.416 in v3 and -0.425 in v1. We fit the file, because it is the likelihood DESI distributes, and we record the difference as a caveat.

For the DR1 comparison we use the DR1 files `desi_2024_gaussian_bao_ALL_GCcomb_{mean,cov}.txt` [2], which have 12 points.

Table 1: DESI DR2 Gaussian BAO likelihood as fitted. The error column is $\sqrt{C_{ii}}$ from the covariance file. Correlations between D_M and D_H in the same bin are included in the fit.

z_{eff}	quantity	value	$\sqrt{C_{ii}}$
0.295	D_V/r_d	7.9417	0.0761
0.510	D_M/r_d	13.5876	0.1684
0.510	D_H/r_d	21.8629	0.4289
0.706	D_M/r_d	17.3507	0.1799
0.706	D_H/r_d	19.4553	0.3339
0.934	D_M/r_d	21.5756	0.1618
0.934	D_H/r_d	17.6415	0.2010
1.321	D_M/r_d	27.6009	0.3246
1.321	D_H/r_d	14.1760	0.2246
1.484	D_M/r_d	30.5119	0.7636
1.484	D_H/r_d	12.8170	0.5180
2.330	D_H/r_d	8.6315	0.1011
2.330	D_M/r_d	38.9890	0.5317

3 Model and method

The primary model, fixed in the preregistration, is flat, with matter and dark energy only:

$$E^2(z) = \frac{H^2(z)}{H_0^2} = \Omega_m(1+z)^3 + (1-\Omega_m)(1+z)^{3(1+w)}, \quad (1)$$

with $w = -1$ for Λ CDM. The distances are

$$D_M = \frac{c}{100 \text{ km s}^{-1} \text{ Mpc}^{-1}} \int_0^z \frac{dz'}{E(z')}, \quad D_H = \frac{c}{100 E(z)} \text{ (same units)},$$

and both are divided by hr_d in Mpc. The isotropic distance is $D_V = [z D_M^2 D_H]^{1/3}$.

The priors are flat, following DESI [2] (Table 2): $\Omega_m \sim U[0.01, 0.99]$, $hr_d \sim U[10, 1000]$ Mpc and $w \sim U[-3, 1]$. DR2 states that its priors “match those given in Table 2 of [38]”. We identify [38] as [2] from context; we did not read the bibliography entry.

We use three estimators:

1. **Primary estimator.** The `emcee` 3.1.6 posterior mean and standard deviation, with 32 walkers and seed 20260927. The distance integral uses 96-point Gauss–Legendre quadrature, which agrees with `scipy.integrate.quad` to 2.4×10^{-15} at the prior corners. The preregistered convergence rule requires $n_{\text{steps}} \geq 50 \tau_{\text{max}}$ and at least 10^4 effective samples.
2. **Grid posterior.** An independent grid posterior over the same priors, with 601^2 points for Λ CDM and 141^3 points for w CDM, spanning ± 10 Fisher standard deviations. The largest marginal mass in any edge cell is 2.7×10^{-8} .
3. **χ^2 minimum.** A Nelder–Mead χ^2 minimum with the Fisher matrix evaluated there. It is used only against DESI’s best-fit χ^2 and in the controls.

All six MCMC runs converged at 20,000 steps. The runs are DR2 and DR1, each in Λ CDM and w CDM, plus the two DR2 radiation variants. The largest autocorrelation time is $\tau_{\text{max}} = 44.3$ and the smallest effective sample size is 14282. Two identical 300-step runs reproduced bit for bit.

4 Preregistration

The preregistration was written before any fit of this problem (revision r2, 2026-09-27 15:55 UTC). It is frozen: no file of it was modified afterwards, and its sha256 is re-checked by every analysis script. It fixes the following:

- the four scored targets;
- two secondary targets: DESI’s best-fit $\chi^2/\text{dof} = 10.2/11$ [1] (Sec. III.C.1) and the Ω_m – hr_d correlation $r = -0.92$;
- the pass rule;
- every control in Table 2.

The pass rule is:

- every target within 0.5 published standard deviations of the published value;

- every width ratio $\sigma_{\text{ours}}/\sigma_{\text{pub}}$ in $[0.8, 1.2]$;
- $|r + 0.92| \leq 0.05$;
- $|\chi_{\text{min}}^2 - 10.2| \leq 0.5$;
- all controls behaving as specified;
- every MCMC run converged;
- all data sha256 digests matching.

The DR2 w CDM value of hr_d is report-only. No BAO-only DR2 value was found in [1] or in [3], so it has no target.

5 Controls

Following this pipeline’s rule that every experiment runs positive and negative controls before its numbers count, the preregistration makes the controls binding: if any positive control fails or any negative control passes, no target agreement may be reported as a success. The script `assemble_fit.py` applies this rule mechanically. The outcomes are listed in Table 2. The noisy mocks in PC3 are 500 draws from $\mathcal{N}(\text{model}(0.2975, 101.54), C_{\text{DR2}})$. Their pulls use the Fisher standard deviation at the truth point.

NC5 passes its preregistered criterion only in aggregate. 6 of the 20 individual scrambled-covariance fits have PTE $> 10^{-3}$ (as seed: PTE, these are 202: 0.022, 208: 0.08, 210: 0.067, 212: 0.0088, 215: 0.0034, 218: 0.007; the largest is 0.080). A scrambled covariance is therefore often still an acceptable fit on its own. This control tests the covariance ordering in aggregate and has limited power for any single realisation.

6 Results

Table 3 shows the four scored targets, and the verdict under the preregistered rule is **PASS**: 14 of 14 criteria hold. The remaining quantities are:

- **Λ CDM secondary targets.** The minimum is $\chi_{\text{min}}^2 = 10.271$ on 11 dof. The Ω_m - hr_d correlation is $r = -0.9230$ from emcee, -0.9239 from the grid and -0.9234 from the Fisher matrix.
- **w CDM.** The minimum is $\chi_{\text{min}}^2 = 9.041$ on 10 dof. The report-only value is $hr_d = 99.83 \pm 1.73$ Mpc.

Figure 1 shows the posteriors.

DR1 instrument check. The same pipeline, run on DR1, gives the following, with the pull relative to [2] eq. (4.1) or (5.1) in parentheses:

- Λ CDM: $\Omega_m = 0.29498$ (pull -0.0013) and $hr_d = 101.887$ Mpc (pull $+0.067$);
- w CDM: $\Omega_m = 0.29305$ (pull $+0.0033$) and $w = -0.9896$ (pull $+0.0025$).

The DR1 w CDM value $hr_d = 101.69$ Mpc (pull -0.0021) is compared against the value in [2], Sec. 6 (printed p. 36): $r_d h = (101.7_{-3.5}^{+2.9})$ Mpc (w CDM). This page was re-fetched in the fix round and matches the literature-review transcription.

Table 2: Preregistered controls. The threshold column gives the preregistered criterion for each control. “As required” means pass for a positive control and fail for a negative one. All values are copied from `controls.json`.

control	threshold	measured	outcome
PC0 DR1 regression	Ω_m, hr_d, χ^2 within $10^{-4}, 10^{-3}, 10^{-3}$ of parent	0.293892, 101.9367, 12.7405	as required
PC1 noiseless Λ CDM mock	$ \Delta\Omega_m < 10^{-4}, \chi^2 < 10^{-4}$	$\Delta\Omega_m = -1.8 \times 10^{-11}$	as required
PC2 noiseless w CDM mock	$ \Delta w < 10^{-3}, \chi^2 < 10^{-4}$	$\Delta w = 1.2 \times 10^{-10}$	as required
PC3 500 noisy mocks	$ \bar{p} < 0.15, 0.88 \leq s_p \leq 1.12, 10.3 \leq \bar{\chi}^2 \leq 11.7$	$\bar{p}_{\Omega_m} = -0.002, s_p = 0.970/0.971, \bar{\chi}^2 = 10.60$	as required
PC4 astropy vs. quad	max rel. diff $< 10^{-5}$	1.6×10^{-15}	as required
NC1 z permutation (all z changed)	must give PTE $< 10^{-3}$	$\chi^2 = 16645$	as required
NC2 $D_M \leftrightarrow D_H$ swap	must give PTE $< 10^{-3}$	$\chi^2 = 36338$	as required
NC3 Einstein-de Sitter	must give PTE $< 10^{-6}$	$\chi^2 = 1420.2, \text{PTE } 6.2 \times 10^{-297}$	as required
NC4 Λ CDM on $w = -0.7$ mock	must give $\chi^2_{\Lambda\text{CDM}} > 9$	$\chi^2 = 11.76$ (w CDM: 2.4×10^{-17})	as required
NC5 20 scrambled covariances	must give median PTE $< 10^{-3}$	median PTE 3.4×10^{-6} ; 6 of 20 single fits have PTE $> 10^{-3}$	as required

Table 3: DR2 BAO-only results against the preregistered targets. The “pull” column is (ours – published)/ σ_{pub} . The “width” column is $\sigma_{\text{ours}}/\sigma_{\text{pub}}$. Values are generated from `fit.json` (generated 2026-09-27 19:14:42 UTC).

parameter	emcee mean $\pm$ sd	grid mean $\pm$ sd	DESI DR2 [1]	pull	width
Λ CDM Ω_m	0.29781 ± 0.00856	0.29783 ± 0.00863	0.2975 ± 0.0086	+0.037	0.995
Λ CDM hr_d [Mpc]	101.530 ± 0.732	101.523 ± 0.736	101.54 ± 0.73	−0.013	1.003
w CDM Ω_m	0.29713 ± 0.00896	0.29714 ± 0.00894	0.2969 ± 0.0089	+0.026	1.007
w CDM w	-0.9164 ± 0.0787	-0.9170 ± 0.0783	-0.916 ± 0.078	−0.005	1.009

DR1→DR2 consistency. From our two posteriors we find $\Delta\Omega_m = +0.00283$ and $\Delta(hr_d) = -0.36$ Mpc. The published difference, derived from the two papers, is $0.2975 - 0.295 = +0.0025$; DESI does not quote it. DR1 is a subset of DR2, so the primary error is the ideal-nesting approximation $\sigma_{\text{nest}} = \sqrt{\sigma_{\text{DR1}}^2 - \sigma_{\text{DR2}}^2}$. With the preregistered $\sigma_{\text{nest}} = 0.01229$ the shift is 0.231σ . With our own $\sigma_{\text{nest}} = 0.01194$ it is 0.237σ . Treating the releases as independent ($\sigma = 0.01700$) gives 0.167σ . All three preregistered shift predictions pass.

Modelling systematics. We ran two checks.

- **CAMB with massless neutrinos.** CAMB 2.0.4, run with massless neutrinos and its own Ω_r , reproduces our quadrature D_M and H to 2.3×10^{-11} relative, for $w = -1$ and $w = -0.9$.
- **CAMB with a 0.06 eV neutrino.** We fitted our primary model, which has no radiation and no neutrino mass, to a noiseless CAMB data vector with one 0.06 eV neutrino. The fitted Ω_m shifts by $+0.032\sigma$ and hr_d by -0.011σ in Λ CDM. In w CDM, w shifts by -0.008σ .

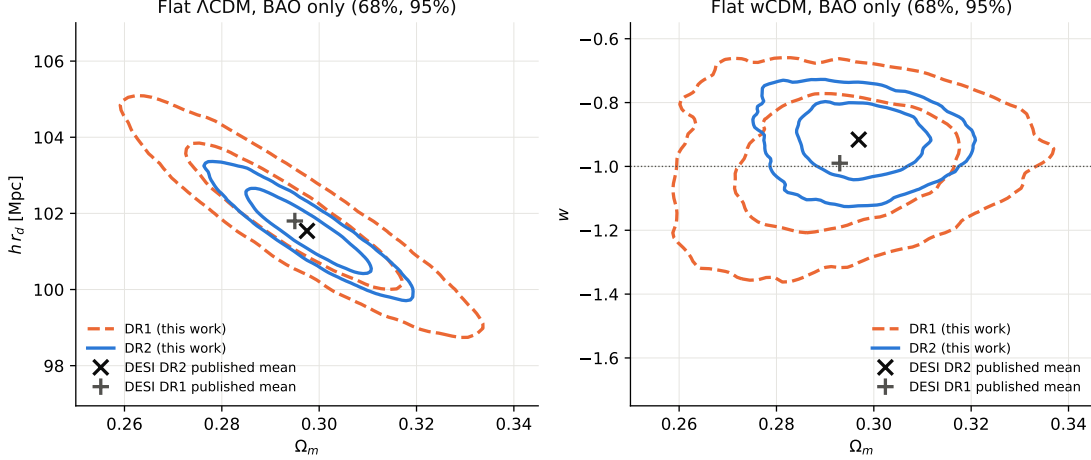


Figure 1: 68% and 95% posterior contours from emcee. Left: flat Λ CDM in $(\Omega_m, h r_d)$. Right: flat w CDM in (Ω_m, w) . Each panel shows DR2 and DR1 (this work) and marks DESI’s published means. Figure produced by `scripts/desi_dr2_bao/make_figure.py` from the chains in `results/desi_dr2_bao/chains/`.

The preregistered radiation variant ($\Omega_r = 9.05 \times 10^{-5}$) shifts Ω_m by -0.046σ , $h r_d$ by $+0.013\sigma$ and w by -0.0016σ . Every shift is far below the 0.3σ flag. CAMB also gives the Planck-2018 best-fit sound horizon $r_{\text{drag}} = 147.1027$ Mpc, matching the orchestrator’s independent positive control.

7 Lean 4 formalization

The file `formal/ANSE/DESI_DR2_wCDM.lean` (sha256 prefix `3b451bd5a93d682c`, toolchain `leanprover/lean4:v4.34.0-rc2`) formalizes the background model that the fit evaluates. It takes `dr2_model.e_of_z` with $\Omega_r = 0$ and the comoving distance $\int_0^z dt/E(t)$. The exponent $3(1+w)$ is a real power (`Real.rpow`), as in `numpy`. Imports are pinned to the locally built Mathlib subset plus `ANSE.BAOFlatLCDM` from the DR1 exercise. The definitions and theorem statements below are extracted verbatim from the file; proofs are elided.

```
noncomputable def E2 (Om w z : ℝ) : ℝ :=
  Om * (1 + z) ^ 3 + (1 - Om) * (1 + z) ^ (3 * (1 + w))

noncomputable def E (Om w z : ℝ) : ℝ := Real.sqrt (E2 Om w z)

noncomputable def Dc (Om w z : ℝ) : ℝ := ∫ t in (0:ℝ)..z, (E Om w t)⁻¹

theorem E2_pos {Om w z : ℝ} (hOm0 : 0 < Om) (hOm1 : Om < 1) (hz : 0 ≤ z) :
  0 < E2 Om w z := ...

theorem E_pos {Om w z : ℝ} (hOm0 : 0 < Om) (hOm1 : Om < 1) (hz : 0 ≤ z) :
  0 < E Om w z := ...

theorem E_w_neg_one (Om z : ℝ) : E Om (-1) z = ANSE.BAOFlatLCDM.E Om z := ...

theorem E_strictMonoOn_of_neg_one_le {Om w : ℝ} (hOm0 : 0 < Om) (hOm1 : Om < 1)
  (hw : -1 ≤ w) : StrictMonoOn (E Om w) (Set.Ici (0 : ℝ)) := ...
```

```

theorem invE_continuousOn {Om w : ℝ} (hOm0 : 0 < Om) (hOm1 : Om < 1) :
  ContinuousOn (fun t => (E Om w t)-1) (Set.Ici (0 : ℝ)) := ...

theorem Dc_strictMonoOn {Om w : ℝ} (hOm0 : 0 < Om) (hOm1 : Om < 1) :
  StrictMonoOn (Dc Om w) (Set.Ici (0 : ℝ)) := ...

```

The statements say the following:

- $E^2 > 0$ and $E > 0$ for $0 < \Omega_m < 1$, $z \geq 0$ and every real w , so the whole prior box is covered, the phantom side included. This is a *preregistration deviation*: the `lean_plan` stated the domain $z > -1$. The delivered $z \geq 0$ is strictly narrower. It still covers every data redshift and the whole prior box. The file was left unchanged in the fix round, so that the statement audit applies to the exact file that was audited.
- At $w = -1$ the w CDM E equals the flat- Λ CDM E of the DR1 formalization, so the two fits are exactly nested.
- E is strictly increasing on $[0, \infty)$ when $w \geq -1$. No claim is made for $w < -1$, where the dark-energy term decreases.
- $1/E$ is continuous on $[0, \infty)$.
- D_C is strictly increasing on $[0, \infty)$ for every real w . This is an interval-integral result, not merely algebra.

The file carries its own `#print axioms` lines. Compiling it with `lake env lean` (return code 0, no errors or warnings, gate run 2026-09-27 17:59:28 UTC) prints exactly:

```

'ANSE.DESIDR2wCDM.E2_pos' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.DESIDR2wCDM.E_pos' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.DESIDR2wCDM.E_w_neg_one' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.DESIDR2wCDM.E_strictMonoOn_of_neg_one_le' depends on axioms: [propext, Classical.
  choice, Quot.sound]
'ANSE.DESIDR2wCDM.invE_continuousOn' depends on axioms: [propext, Classical.choice, Quot.
  sound]
'ANSE.DESIDR2wCDM.Dc_strictMonoOn' depends on axioms: [propext, Classical.choice, Quot.
  sound]

```

There is no `sorryAx` and no axiom outside `{propext, Classical.choice, Quot.sound}`. The Elenchus scanner `elenchus_check.py` reports “no findings”. To check that the scanner can detect a problem in this file, we built a copy in which the proof of `E_w_neg_one` is replaced by `sorry`. Elenchus flags that copy `INHERITED_SORRY` and exits with return code 1 (`results/desi_dr2_bao/lean_gate_negative_control.json`).

Claim ledger. The ledger registers 25 claims in Elenchus’s tier-capped format. Six are Tier A (the theorems above) and eight are Tier B (our computations). Ten are Tier L: five published values and priors, and five comparisons against them, which rest on citations. One is a Tier C argument that the Lean definitions are faithful to the code.

Before the statement audit, the gate reported 0 blocking findings and 6 flags, all `LEDGER_UNAUDITED_TIER_A`, with return code 1. A workflow referee then audited the six Lean statements against `dr2_model.py`. That referee is an automated model, not a person. It judged all six faithful (`results/desi_dr2_bao/referee_report.json`). Its verdicts are recorded as audit

objects on 6 of the 6 Tier A rows. Each object names the auditor as a model, and each is bound to the sha256 of the audited Lean file.

With these audits the gate reports 0 blocking findings and 0 flags, and its return code is 0. Elenchus’s flag text asks for a *person*’s certification, so this return code means “model-referee audited”. A human statement audit is still outstanding. Three mutations of the ledger were each caught as a block: a tier inversion, a tampered evidence blob, and a kind over-claim.

8 Limitations

- **A reproduction, not a new constraint.** We use DESI’s likelihood, priors and estimator. Agreement is expected by construction. The value of the exercise lies in the independent code path, the preregistration and the controls.
- **Different sampler and theory code.** We use `emcee` and analytic quadrature, while DESI uses Cobaya Metropolis–Hastings with CAMB. The physics missing from our primary model (radiation and a 0.06 eV neutrino) moves no parameter by more than 0.046σ (CAMB check and radiation variant together).
- **Covariance-file correlations.** The file’s D_M – D_H correlations differ from Table IV of [1] for LRG3+ELG1 and ELG2. We fit the file.
- **Prior source.** The source of the priors is inferred: we identified “[38]” in [1] from context.
- **Table V version.** The w CDM row of Table V was read in v3 only. The Λ CDM targets are identical in v1 and v3.
- **Nesting approximation.** σ_{nest} is an ideal-nesting approximation. DESI’s own DR1/DR2 correlation model is $C = \sqrt{N_{\text{DR1}}/N_{\text{DR2}}}$ per tracer (0.61 for $\text{Ly}\alpha$), which we did not reproduce.
- w CDM $h r_d$. It is report-only, with no published target.
- **Lean scope and audit.**
 - The Lean theorems are facts about the model, not about the data or the fit.
 - The Tier A rows were audited for statement faithfulness by an automated model referee only, not by a person.
 - $E^2 > 0$ and $E > 0$ are proved for $z \geq 0$, not for the preregistered $z > -1$ (a preregistration deviation, see Sec. 7).
 - The radiation variant is not formalized.
 - The gate that ran is `scripts/desi_dr2_bao/lean_gate.py`. It performs the same compile, `#print axioms`, `sorryAx` and `whitelist` checks. The preregistration names `anse/formal/lean_runner.py`, which was not used because it needs a `lake build` of the module.
- **Script edit after the runs (closed).** After the first chains were written, `mcmc_fit.py` received an annotation-only edit (a return type). In the fix round all six chains and both grids were regenerated from the current `mcmc_fit.py` and `grid_posterior.py`. Every chain is bit-identical to its predecessor (`np.array_equal`), and both summaries are identical apart from timestamps (`results/desi_dr2_bao/fixround_rerun_check.json`).

Acknowledgments

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References

- [1] DESI Collaboration, “DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints,” arXiv:2503.14738 (v3, 9 Oct 2025; v1 checked for eq. 17 and Table IV).
- [2] DESI Collaboration, “DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations,” arXiv:2404.03002 (v3).
- [3] K. Lodha et al., DESI DR2 extended dark-energy analysis, arXiv:2503.14743 (v2); title not transcribed in our literature review. Consulted only to confirm that no BAO-only DR2 w CDM $h r_d$ value is given in the pages returned.